

Document Question Answering System

Repository

Role fit: Machine Learning, AI Engineering, RAG, Python

Skills: Python, RAG, LangChain, FAISS, Streamlit, Google Embeddings, Groq, Llama 3

Streamlit retrieval-augmented generation application for answering questions from PDF document collections.

- Built a Streamlit document-question-answering application that ingests local PDF collections and answers user queries from retrieved context.
- Implemented a LangChain ingestion pipeline with PDF loading, recursive 1,000-character chunks with 200-character overlap, Google embeddings, and a FAISS vector store.
- Connected a FAISS retriever to a context-constrained document chain and Groq-hosted Llama 3 generation for grounded question answering.
- Persisted embeddings in Streamlit session state so users can ingest once, ask multiple questions, and inspect response timing in the same interactive workflow.